

Exercise 5: Triggers

Aim

To implement MySQL triggers that automatically execute during INSERT and UPDATE operations to maintain data consistency, audit logs, and enforce banking business rules.

--**Question:** Write a trigger **UpdateCustomerLastModified** that updates the LastModified column of the Customers table to the current date whenever a customer's record is updated.

```
ALTER TABLE Customer
```

```
ADD LastModified DATE;
```

```
DELIMITER $$
```

```
CREATE TRIGGER UpdateCustomerLastModified
```

```
BEFORE UPDATE
```

```
ON Customer
```

```
FOR EACH ROW
```

```
BEGIN
```

```
SET NEW.LastModified = CURDATE();
```

```
END $$
```

```
DELIMITER ;
```

```
UPDATE Customer
SET Balance = 25000
WHERE CustomerID = 101;
```

```
SELECT CustomerID,
       CustomerName,
       Balance,
       LastModified
FROM Customer;
```

	CustomerID	CustomerName	Balance	LastModified
▶	101	Krishna	25000.00	2026-06-29
	102	Rahul	8000.00	NULL
	103	Priya	20000.00	NULL
	104	Arun	12000.00	NULL
	105	Sneha	5000.00	NULL
	106	Karan	12000.00	NULL
✱	NULL	NULL	NULL	NULL

--**Question:** Write a trigger **LogTransaction** that inserts a record into an AuditLog table whenever a transaction is inserted into the Transactions table.

```
CREATE TABLE Transactions(
    TransactionID INT PRIMARY KEY,
    AccountID INT,
    TransactionType VARCHAR(20),
    Amount DECIMAL(10,2),
    FOREIGN KEY(AccountID) REFERENCES Account(AccountID)
);
```

```
CREATE TABLE AuditLog(
    LogID INT AUTO_INCREMENT PRIMARY KEY,
    TransactionID INT,
    ActionDate DATETIME,
    Description VARCHAR(100)
);
```

DELIMITER \$\$

CREATE TRIGGER LogTransaction

AFTER INSERT
ON Transactions

FOR EACH ROW

BEGIN

INSERT INTO AuditLog

(
 TransactionID,
 ActionDate,
 Description
)

VALUES

(
 NEW.TransactionID,
 NOW(),
 'Transaction Inserted'
);

END \$\$

DELIMITER ;

INSERT INTO Transactions

VALUES

(1,1,'Deposit',5000);

SELECT * FROM AuditLog;

	LogID	TransactionID	ActionDate	Description
▶	1	1	2026-06-30 00:03:33	Transaction Inserted
•	NULL	NULL	NULL	NULL

--**Question:** Write a trigger **CheckTransactionRules** that ensures withdrawals do not exceed the balance and deposits are positive before inserting a record into the Transactions table.

DELIMITER \$\$

CREATE TRIGGER CheckTransactionRules

BEFORE INSERT

ON Transactions

FOR EACH ROW

BEGIN

DECLARE CurrentBalance DECIMAL(10,2);

-- Check Deposit

IF NEW.TransactionType='Deposit'

AND NEW.Amount<=0 THEN

SIGNAL SQLSTATE '45000'

SET MESSAGE_TEXT='Deposit amount must be positive';

END IF;

-- Check Withdrawal

IF NEW.TransactionType='Withdrawal' THEN

SELECT Balance

```
INTO CurrentBalance
FROM Account
WHERE AccountID=NEW.AccountID;

IF NEW.Amount>CurrentBalance THEN

SIGNAL SQLSTATE '45000'
SET MESSAGE_TEXT='Insufficient Balance';

END IF;

END IF;

END $$

DELIMITER ;

INSERT INTO Transactions
VALUES
(2,1,'Deposit',-500);
```

❌ 11 00:07:50 INSERT INTO Transactions VALUES (2,1,'Deposit',-500)
Error Code: 1644. Deposit amount must be positive

```
INSERT INTO Transactions
VALUES
(3,1,'Withdrawal',100000);
```

❌ 12 00:09:38 INSERT INTO Transactions VALUES (3,1,'Withdrawal',100000) |
Error Code: 1644. Insufficient Balance

```
INSERT INTO Transactions
```

```
VALUES
```

```
(4,1,'Withdrawal',5000);
```

```
SELECT * FROM Transactions;
```

	TransactionID	AccountID	TransactionType	Amount
▶	1	1	Deposit	5000.00
	4	1	Withdrawal	5000.00
✱	NULL	NULL	NULL	NULL